THE EXACT-SUPPORT SHEAF AS RIGHT ADJOINT TO PUSHFORWARD AND ITS SECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A75 (sources/stacks-duality/pinned-original-source.tex line 2283).

Allowed standard prerequisites (editorial).

- Adjunction between tensor product and internal hom. Modules, Lemma labelled modules-lemma-adjoint-tensor-restrict, cited at proof lines 2300-2304. Gives the first equality of the displayed chain.
- Full faithfulness of pushforward along a closed immersion. Modules, Lemma labelled modules-lemma-i-star-equivalence, cited at proof lines 2304-2306. Gives the second equality of the displayed chain.

Source reference. Original lemma 0A6F: sources/stacks-chow/tags.txt, source lines 12843–12843. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2304 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08KS: sources/stacks-chow/tags.txt, source lines 10895–10895. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2306 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *With notation as above. The functor $\mathcal{H}om(\mathcal{O}_Z, -)$ is a right adjoint to the functor $i_* : Mod(\mathcal{O}_Z) \to Mod(\mathcal{O}_X)$. For $V \subset Z$ open we have*

$$\Gamma(V, \mathcal{H}om(\mathcal{O}_Z, \mathcal{F})) = \{s \in \Gamma(U, \mathcal{F}) \mid \mathcal{I}s = 0\}$$

where $U \subset X$ is an open whose intersection with Z is V.

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Proof. Let $\mathcal{G}$ be a sheaf of $\mathcal{O}_Z$-modules. Then

$$\mathrm{Hom}_{\mathcal{O}_X}(i_*\mathcal{G}, \mathcal{F}) = \mathrm{Hom}_{i_*\mathcal{O}_Z}(i_*\mathcal{G}, \mathcal{H}om_{\mathcal{O}_X}(i_*\mathcal{O}_Z, \mathcal{F})) = \mathrm{Hom}_{\mathcal{O}_Z}(\mathcal{G}, \mathcal{H}om(\mathcal{O}_Z, \mathcal{F}))$$

The first equality by Modules, Lemma 0A6F and the second by the fully faithfulness of i_*, see Modules, Lemma 08KS. The description of sections is left to the reader. □